CARBOplatin (AUC 2) Weekly with Radiotherapy (RT)

INDICATIONS FOR USE:

INDICATION	ICD10	Regimen Code	HSE approved reimbursement status*
Chemoradiation treatment for stage III and IV locally advanced head and neck squamous cell carcinoma in patients where CISplatin is contraindicated or not tolerated[i]	C11	00419a	Hospital
Chemoradiation treatment in locally advanced cervical cancer when CISplatin is contraindicated or not tolerated[i]	C53	00419b	Hospital

* This applies to post 2012 indications

TREATMENT:

The starting dose of the drugs detailed below may be adjusted downward by the prescribing clinician, using their independent medical judgement, to consider each patient's individual clinical circumstances.

Head and neck squamous cell Carcinoma
CARBOplatin is administered concomitantly with radiotherapy on Day 1 of each cycle.

This treatment is followed by adjuvant chemotherapy beginning 4 weeks after the end of radiation therapy. **Refer to NCCP SACT Regimen 00552: CARBOplatin (AUC5) and 5-Fluorouracil 1000mg/m^2/day Therapy – 28-day cycle.**

Cervical Cancer
CARBOplatin is administered concomitantly with radiotherapy on Day 1 of each cycle.

Cycle frequency and duration:

Each cycle is 7 days. Treatment should continue for a total of 6 cycles unless the patient experiences disease progression or unacceptable toxicity.
CARBOplatin is only to be administered concurrently with radiotherapy. Treatment with CARBOplatin should be discontinued if radiotherapy is truncated.

Facilities to treat anaphylaxis MUST be present when systemic anti-cancer therapy (SACT) is administered.

Day	Drug	Dose	Route	Diluent and Rate	Cycle
1	CARBOplatin	AUC 2	IV infusion	250mL glucose 5% over 30 minutes	1-6

Note: Administration volumes and fluids have been standardised to facilitate electronic prescribing system builds.

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CARBOplatin dose:

The dose in mg of CARBOplatin to be administered is calculated as follows:

Dose (mg) = target AUC (mg/mL x minute) x (GFR mL/minute +25)

- **Measured GFR** (e.g. nuclear renogram) is preferred whenever feasible
- **Estimation of GFR** (eGFR) can be done by using the Wright formula or using the Cockroft and Gault formula to measure creatinine clearance.
- The GFR used to calculate the AUC dosing should not exceed 125 mL/minute.
    - For obese patients and those with a low serum creatinine, for example, due to low body weight or post-operative asthenia, estimation using formulae may not give accurate results; measured GFR is recommended.
        - Where obesity (body mass index [BMI] ≥ 30 kg/m^2) or overweight (BMI 25-29.9) is likely to lead to an overestimate of GFR and isotope GFR is not available the use of the adjusted ideal body weight in the Cockcroft and Gault formula may be considered.
        - Where serum creatinine is less than 63 micromol/L, the use of a creatinine value of 62 micromol/L or a steady pre-operative creatinine value may be considered
    - These comments do not substitute for the clinical judgement of a physician experienced in prescription of CARBOplatin.

WRIGHT FORMULA

There are two versions of the formula depending on how serum creatinine values are obtained, by the kinetic Jaffe method or the enzymatic method. The formula can be further adapted if covariant creatine kinase (CK) values are available (not shown).

1. *SCr measured using enzymatic assay.*

$$\text{GFR (mL/minute)} = \frac{(6230 - 32.8 \times \text{Age}) \times \text{BSA} \times (1 - 0.23 \times \text{Sex})}{\text{SCr (micromol/minute)}}$$

2. *SCr measured using Jaffe assay*

$$\text{GFR (mL/minute)} = \frac{(6580 - 38.8 \times \text{Age}) \times \text{BSA} \times (1 - 0.168 \times \textit{Sex})}{\text{SCr (micromol /minute)}}$$

Key: Sex = 1 if female, 0 if male; Age in years; BSA= DuBois BSA

COCKCROFT-GAULT FORMULA

$$\text{GFR (mL/minute)} = \frac{\text{S} \times (140 - \text{age in years}) \times \text{wt (kg)}}{\text{serum creatinine (micromol/L)}}$$

S= 1.04 for females and 1.23 for males

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ELIGIBILITY:

- Indications as above
- ECOG status 0-2

EXCLUSIONS:

- Hypersensitivity to CARBOplatin or any of the excipients
    - If it is felt that the patient may have a major clinical benefit from CARBOplatin, it may in exceptional circumstances be feasible to rechallenge a patient with a prior mild hypersensitivity reaction e.g. using a desensitisation protocol, but only with immunology advice, premedication as advised, and a desensitisation protocol under carefully controlled conditions with resuscitation facilities available and medical and/or ITU/ HDU supervision
- Pregnancy
- Breastfeeding

PRESCRIPTIVE AUTHORITY:

The treatment plan must be initiated by a Consultant Medical Oncologist.

TESTS:

Baseline tests:

- FBC, renal and liver profile
- Isotope GFR measurement (preferred) or GFR / CrCl estimation
- Audiology if clinically indicated
- Assessment of peripheral neuropathy status as clinically indicated prior to starting treatment

Regular tests:

- FBC, renal and liver profile before each cycle.

Disease monitoring:
Disease monitoring should be in line with the patient's treatment plan and any other test/s as directed by the supervising Consultant.

DOSE MODIFICATIONS:

- Any dose modification should be discussed with a Consultant

Haematological:

Table 1: Dose modification of CARBOplatin in haematological toxicity

ANC (x 10^9/L) (on day of chemotherapy)		Platelets (x 10^9/L) (at any stage during cycle)	Dose Modification
≥ 1	and	≥ 100	100% Dose
0.5-0.99	or	50-99	Delay treatment until recovery
< 0.5	or	< 50	Delay treatment until recovery and consider reducing CARBOplatin by 25% for subsequent cycles
Febrile neutropenia			

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Renal and Hepatic Impairment:

Table 2: Dose modification of CARBOplatin in renal and hepatic impairment

Drug	Renal Impairment	Hepatic Impairment
CARBOplatin	See note below*	No dose modification required

- Patients with creatinine clearance values of < 60mL/minute are at greater risk of developing myelosuppression.
- If GFR between 20 to ≤ 30mL/minute, CARBOplatin should be administered with extreme caution
- If GFR ≤ 20mL/minute, CARBOplatin should not be administered at all
- If Cockcroft & Gault or Wright formula are used, the dose should be calculated as required on each cycle based on a serum creatinine obtained within 48 hours of drug administration.
- If isotope GFR is used, the dose can remain the same provided the serum creatinine is ≤ 110% of its value at the time of the isotope measurement. If the serum creatinine increases, consideration should be given to remeasuring the GFR or to estimating it using Cockcroft & Gault or Wright formulae, taking care this does result in a CARBOplatin dose reduction.

SUPPORTIVE CARE:

EMETOGENIC POTENTIAL:

- As outlined in NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting - Available on the NCCP website

CARBOplatin: Moderate (**Refer to local policy)**

For information:
Within NCIS regimens, antiemetics have been standardised by the Medical Oncologists and Haemato-oncologists. Information is available in the following documents:

- NCCP Supportive Care Antiemetic Medicines for Inclusion in NCIS (Medical Oncology) - Available on the NCCP website
- NCCP Supportive Care Antiemetic Medicines for Inclusion in NCIS (Haemato-oncology) - Available on the NCCP website

PREMEDICATIONS: None usually required

OTHER SUPPORTIVE CARE: No specific recommendations.

ADVERSE EFFECTS:

- Please refer to the relevant Summary of Product Characteristics (SmPC) for details.
- Please note the NCCP National SACT regimens do not provide information regarding black triangle (▼) status. Please refer to the relevant Summary of Product Characteristics (SmPC) for this information.

REGIMEN SPECIFIC COMPLICATIONS:

- Please refer to the relevant Summary of Product Characteristics (SmPC) for details

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- **Hypersensitivity**: Reactions to CARBOplatin may develop in patients who have been previously exposed to platinum therapy. However allergic reactions have been observed upon initial exposure to CARBOplatin.
- **Neurotoxicity and ototoxicity:** Neurological evaluation and an assessment of hearing should be performed on a regular basis, especially in patients receiving high dose CARBOplatin. Neurotoxicity, such as parasthesia, decreased deep tendon reflexes, and ototoxicity are more likely seen in patients previously treated with CISplatin, other platinum treatments and other ototoxic agents. Frequency of neurologic toxicity is also increased in patients older than 65 years.

DRUG INTERACTIONS:

- Current SmPC and drug interaction databases should be consulted for information.

REFERENCES:

1. Chitapanarux I, Lorvidhaya , Kamnerdsupaphon P. et al. Chemoradiation comparing cisplatin versus carboplatin in locally advanced nasopharyngeal cancer: randomised, non-inferiority, open trial. Eur J Cancer 2007;43(9):1399-1406.
2. Katanyoo k, Tangjitgamol S, et al. Treatment outcomes of concurrent weekly carboplatin with radiation therapy in locally advanced cervical cancer patients. J Gynae Onc 2011; 123: 571-576
3. Ekhart C, Rodenhuis S et al. Carboplatin dosing in overweight and obese patients with normal renal function, does weight matter? Cancer Chemother Pharmacol 2009;64:115-122 https://link.springer.com/article/10.1007%2Fs00280-008-0856-x
4. Wright JG, Boddy AV, et al, Estimation of glomerular filtration rate in cancer patients. British Journal of Cancer 2001; 84(4):452-459
5. Appropriate chemotherapy dosing for obese adult patients with cancer: American Society of Clinical Oncology Clinical Practice Guideline. J Clin Oncol 2012; 30 (13) 1553-1561.
6. NCCN CARBOplatin dosing in adults https://www.nccn.org/docs/default-source/clinical/order-templates/appendix_b.pdf?sfvrsn=6286822e_6
7. Krens S D, Lassche,Jansman G F G A, et al. Dose recommendations for anticancer drugs in patients with renal or hepatic impairment. Lancet Onco/2019; 20:e201-08. https://doi.org/10.1016/S1470-2045(19)30145-7
8. NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting. V6 2025. Available at: https://healthservice.hse.ie/documents/7152/Classification_Document_for_Systemic_Anti-_Cancer_Therapy_SACT_Induced_Nausea__pWgDCrm.pdf
9. CARBOplatin (Teva) 10mg/mL Concentrate for Solution for Infusion. Summary of Product Characteristics HPRA – Last updated 14/03/2024. Accessed 07/05/2026. Available at: https://assets.hpra.ie/products/Human/20850/Licence_PA0749-004-001_14032024104516.pdf

From 04/08/2026, this regimen will be reviewed on an 'As Required' basis. Please refer to the version control box for information, and the footer for date of last review. If you wish to suggest any updates to this regimen, please contact oncologydrugs@cancercontrol.ie

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Version	Date	Amendment	Approved By
1	08/05/2017		Prof Maccon Keane
2	15/05/2019	Inclusion of unlicensed status. Standardisation of treatment table. Update of drug interactions	Prof Maccon Keane
3	03/02/2021	Reviewed. New Indication added.	Prof Maccon Keane
4	29/08/2022	Updated CARBOplatin infusion time. Updated standard wording for CARBOplatin dosing and creatinine value. Updated baseline tests. Updated dose modification of CARBOplatin in haematological toxicity.	Prof Maccon Keane
5	04/08/2026	Regimen reviewed. Updated exclusion criteria. Updated baseline and regular testing. Updated regimen in line with NCCP standardisation.	Prof Maccon Keane
6	23/09/2026	Indication wording updated for indication 00419a	Dr Colm Mac Eochagáin

Comments and feedback welcome at oncologydrugs@cancercontrol.ie.

[i]This is an unlicensed posology and indication for the use of CARBOplatin in Ireland. Patients should be informed of this and consented to treatment in line with the hospital's policy on the use of unlicensed medication and unlicensed or "off label" indications. Prescribers should be fully aware of their responsibility in communicating any relevant information to the patient and also ensuring that the unlicensed or "off label" indication has been acknowledged by the hospital's Drugs and Therapeutics Committee, or equivalent, in line with hospital policy.

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